Final Project Report: Knowledge Graph Construction, Alignment, Reasoning, KGE & Semantic RAG

Software Engineering & Data Science Project

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Abstract

This report consolidates the findings of an end-to-end Knowledge Engineering pipeline. It details the journey from unstructured web data acquisition and Information Extraction (IE) to Knowledge Base (KB) construction, alignment, and topological expansion. It further explores explicit rule-based reasoning (SWRL), implicit continuous reasoning via Knowledge Graph Embeddings (KGE), and concludes with the implementation of a Semantic Retrieval-Augmented Generation (RAG) system powered by a local Small Language Model (SLM).

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1. Data Acquisition & Information Extraction (IE)

1.1. Methodology

Our data acquisition pipeline leveraged `trafilatura` to process a set of 5 initial seed URLs focusing on AI research (e.g., UKRI, NSF, The Guardian). `trafilatura` proved highly efficient at boilerplate removal, cleanly isolating the main textual content from UI noise. To ensure data quality and sufficient text volume, we implemented a filtration step rejecting any page containing fewer than 500 words. All 5 URLs successfully passed this validation.

For the Information Extraction phase, we utilized the `en_core_web_trf` transformer model via `spaCy`. This model performed Named Entity Recognition (NER) to classify nodes (ORG, PERSON, GPE, DATE) and Dependency Parsing to extract edges (relations) by linking subjects and objects connected by verbal predicates. This pipeline successfully yielded a foundational graph of 239 entities and 36 extracted relations.

1.2. Ambiguity Analysis

During the NLP extraction, we observed common types of entity ambiguity inherent to open-text parsing:

- **Contextual Ambiguity / Polysemy:** Acronyms (e.g., "AI") or broad organizational terms can be misclassified depending on the surrounding sentence structure.
- **Type Overlap:** Entities such as specific institutions (e.g., "Caltech") can represent a `PERSON` (due to noun-adjunct parsing errors), a `GPE` (location), or an `ORG` (university).
- **Metonymy:** Journalistic texts frequently use a `GPE` to represent a government or `ORG` (e.g., "Europe regulates AI"), leading the NLP model to tag it strictly geographically rather than functionally.

2. Knowledge Base Construction, Alignment & Expansion

2.1. Alignment Strategy

To align our private Knowledge Base (496 initial triplets) with an open-world graph, we utilized the Wikidata API (`wbsearchentities` action). The pipeline queried extracted `ORG`, `PERSON`, and `GPE` entities, successfully returning 97 confident alignments. Matched entities were linked to our private URIs using the `owl:sameAs` predicate. Unmatched entities were classified dynamically, and their ontological definitions (e.g., `rdfs:subClassOf`) were serialized into a separate `ontology.ttl` file.

2.2. Expansion Strategy & Topological Balancing

Expanding the graph via Wikidata immediately presented a topological conflict with the strict assignment constraints (Triplets: 50k-200k, Entities: 5k-30k, Relations: 50-200).

Our initial 1-Hop and 2-Hop SPARQL queries extracted a massive 115,155 raw triplets. However, Wikidata's structure resembles a "star graph" with extreme sparsity. When we filtered the graph to retain only top relations, the entity count bloated to 56,221 (violating the 30,000 maximum constraint). Simply pruning these disconnected "leaf" entities mathematically forced the triplet count below the 50,000 minimum limit.

To overcome this mathematical impossibility, we employed a highly effective knowledge engineering technique: **Inverse Relation Injection**. First, we strictly pruned the graph's fringe entities until a dense core of 20,223 entities remained. Then, for the remaining triplets, we programmatically generated inverse relations (`s p o` → `o p_inverse s`). This strategy successfully doubled the graph's density without introducing a single new entity.

2.3. Final KB Statistics

By applying the topological pruning and inverse injection, the final dataset (`expanded_kb.nt`) strictly and safely conforms to all required parameters for robust Knowledge Graph Embedding:

- **Number of triplets:** 70,332 (Compliant with 50k-200k constraint)
- **Number of entities:** 20,223 (Compliant with 5k-30k constraint)
- **Number of relations:** 200 (Compliant with 50-200 constraint)

3. Reasoning (SWRL)

This phase evaluated reasoning over a Knowledge Base using explicit logic. Using `owlready2` and the Pellet reasoner, we executed Semantic Web Rule Language (SWRL) rules on a family ontology. The system successfully inferred and re-parented `family.Alice` into the `oldPerson` class based on numerical data properties (execution time: ∼2.05s). This deterministic reasoning serves as an explicit Open-World Assumption (OWA) baseline to compare against the probabilistic, implicit reasoning models generated in the subsequent embedding phase.

4. Knowledge Graph Embeddings (KGE)

4.1. Methodology & Hyperparameters

We processed the expanded KB from the previous phase using `PyKEEN`. Literals were filtered out, and a strict K-core decomposition (K=2) was applied to prevent entity leakage. The dataset was split into Training (80%), Validation (10%), and Testing (10%).

Both KGE models (TransE and ComplEx) were evaluated using a unified configuration to ensure a fair baseline comparison: **Embedding Dimension:** 100, **Optimizer:** Adam (Learning Rate: 0.01), **Batch Size:** 256, **Number of Epochs:** 50, **Loss Function:** Margin Ranking Loss.

4.2. Experimental Results

Metric	TransE	ComplEx
MRR	**0.0234**	0.0061
Hits@10	**0.0452**	0.0114

Table 1: Model Comparison (Full Dataset Split)

Subsample Size	MRR	Observation on Link Prediction
Small Sample (20k)	**0.0343**	Highest MRR. Contiguous sampling preserved local dense clusters.
Medium Sample (50k)	0.0189	Decrease in MRR due to increasing structural sparsity.
Full Dataset (∼70k)	0.0234	Slight recovery, but impacted by algorithmic data pruning (see Section 6).

Table 2: KB Size Sensitivity (TransE)

4.3. Analysis

- **Model Comparison:** TransE drastically outperformed ComplEx. Because our graph possesses only 200 relations mapped across 20,000 entities, the feature space is extremely sparse. ComplEx, which uses complex-valued embeddings (doubling the parameter space), suffered from severe overfitting. TransE's simpler translational geometry was far more robust to this sparsity.
- **Nearest Neighbors:** Cosine Similarity on entity embeddings successfully grouped semantically related nodes. For example, the nearest neighbors for `http://americanart.si.edu/` included relevant Wikidata entities (e.g., Q1201886, Q134661623) with tight cosine distances (∼0.26 to 0.28), demonstrating the model's ability to cluster cultural institutions effectively.
- **Rule vs. Vector Reasoning:** We tested if vector arithmetic could approximate relational composition (`rank` + `type` ≈ `isDefinedBy`). Calculating the cosine similarity yielded a score of 0.1718. While positive, the relatively low similarity indicates that continuous embeddings capture broad semantic nuances but struggle to enforce strict logical compositions perfectly, unlike deterministic SWRL.

5. Retrieval-Augmented Generation (RAG) over RDF/SPARQL

5.1. Architecture & Methodology

The final objective was bridging unstructured natural language and our structured graph through a Semantic RAG pipeline. Using `llama3.2:1b` locally via `Ollama`, the system architecture was designed around a resilient execution loop:

1. **Schema Extraction:** Upon initialization, the script parses the graph and extracts the top 50 classes and predicates.
2. **Contextual Prompting:** The user's query is merged with the schema to restrict the LLM to valid URIs.

3. **Execution:** The generated query is parsed and executed against the graph by `rdflib`.
4. **Self-Repair Mechanism:** If a syntax error is raised, the script captures the traceback and feeds it back to the LLM for a second "repair" attempt.

5.2. Experimental Results & Observations

We conducted a series of stress tests on the chatbot. The results yielded profound insights into the behavior of 1-Billion-parameter models when tasked with rigid code generation.

- **The Baseline Necessity (Hallucinations):** When asked questions without RAG access, the Baseline model consistently hallucinated. For instance, when asked to find organizations, it suggested writing queries in a non-existent language ("NTP"). When asked to show 10 triples, it confidently generated geometric text about "triangles with vertices." This proves that without the RAG pipeline, the LLM is completely blind to domain-specific structured data.
- **Structural & Syntax Hallucinations (SPARQL Generation):** Despite successfully injecting the schema, `llama3.2:1b` demonstrated severe limitations in understanding SPARQL syntax:
  - *The XML Confusion:* When presented with long URIs from the schema, the model repeatedly associated them with web metadata protocols. Instead of generating a `SELECT` statement, it generated massive, fabricated XML/RDF documents.
  - *The Imperative Confusion:* In our most restrictive test, we explicitly commanded: *"Only use the SPARQL keyword SELECT. Do not use XML."* The model bypassed the XML issue but invented a hybrid syntax using `FOR` loops and `START-FF` commands, which do not exist in SPARQL.
- **The Self-Repair Loop in Action:** From an engineering standpoint, the pipeline performed flawlessly. In every instance of syntax failure (e.g., *Expected SelectQuery, found 'FOR'*), the script successfully caught the exception and triggered the `[Repaired via LLM?] True` mechanism. However, because the 1B model's context window was already saturated by its initial hallucination, it was structurally unable to fix its own code.

6. Critical Reflection & Conclusions

The experiments clearly separate the success of the software engineering from the limitations of the AI model. Consolidating the pipeline reveals key takeaways:

- **Scalability in Data Acquisition:** If the objective shifted from crawling 5 pages to 10,000 pages, our synchronous Python script would become a severe bottleneck. To scale effectively, we would need to transition to asynchronous scraping (e.g., `asyncio` with `httpx` or `Scrapy`), implement proxy rotation, and replace local file storage with distributed databases (MongoDB/Neo4j).
- **The PyKEEN Inverse Relation Anomaly:** A fascinating algorithmic interaction occurred between our TD4 topological design and PyKEEN's default behavior. In

TD4, we explicitly injected `_inverse` relations to densify the graph. However, during the full graph PyKEEN initialization, the library detected these suffixes and automatically purged them to prevent data leakage (`keeping 0.5 triples`). This halved the graph's structural density under the hood, dropping it from 70,000 to ∼35,000 active triples. This anomaly perfectly explains our Size Sensitivity results: the 20k sample was extracted contiguously, preserving local density and bypassing the filter, thereby scoring a higher MRR (0.0343) than the automatically-pruned full graph (0.0234).

- **SLM Capabilities & Context Saturation:** The Semantic RAG module proved highly robust, handling context extraction and error-catching autonomously. However, translating Natural Language to strict SPARQL using long URIs imposes a high cognitive load (Context Saturation). A model with only 1 Billion parameters lacks the necessary attention mechanisms to separate the schema URIs from rigid SPARQL syntax rules.

Final Verdict: Explicit logic (SWRL) offers absolute precision but cannot scale across noisy web data. Knowledge Graph Embeddings (TransE) scale exceptionally well and excel at clustering, but their predictive power (MRR) is extremely sensitive to graph density and automated pruning (as seen with ComplEx's overfitting). Ultimately, Semantic RAG solves the accessibility issue of knowledge graphs, but to achieve production-level query accuracy, the exact same Python architecture should be maintained while replacing the `llama3.2:1b` engine with a higher-capacity model (e.g., Llama 3 8B) capable of adhering strictly to logical constraints.